

Video Transcripts

Week 3.2: Advanced Strategies and Reliability for Business Outputs

Video 1: Need for Advanced Prompting

Over the past year, organisations have moved beyond using generative AI for simple tasks. The real shift in the market is its use in decision support, where AI is expected to interpret data, evaluate options, and guide action. Consider a common scenario in many organisations today.

A business unit leader is reviewing a decline in quarterly performance in one of their key regions. The situation is not straightforward. Sales have dropped, but marketing spend has increased. Customer acquisition is steady, yet retention is weakening. At the same time, the operational costs in the region have gone up. The question is not just what happened, but what should we do next? The leader turns to an AI assistant and provides all relevant inputs like sales data, trends, marketing metrics, customer churn indicators, and cost movements, and asks for an analysis and recommendation.

At the first glance, this seems like a well-structured prompt, but the output often falls short. The AI may summarise each data point correctly but fail to connect them meaningfully. It may suggest generic actions like improve retention or optimise marketing spend without identifying the underlying trade-offs or prioritising interventions. This is the limitation of simple prompting. It treats the problem as a single-step task, i.e. analyse and recommend, when in reality the situation requires multi-step reasoning.

The problem needs to be broken down. First, identifying the patterns. Then, diagnosing root causes. Then, evaluating possible actions and finally recommending a course of action. Without that structure, the AI tends to jump directly to an answer, often skipping the reasoning that makes the answer reliable. There is also a consistency challenge.

In business environments, the same prompt may produce different levels of depth or quality across attempts. When decisions involve revenue, cost, or customer impact, this variability becomes a serious concern. As organisations scale the use of AI across teams, this issue becomes more visible. It is no longer enough for AI to generate responses. It must produce outputs that are consistent, explainable, and aligned with decision needs. This is where advanced prompting becomes essential.

Instead of asking the AI to solve the entire problem in one step, advanced prompting guides it through a structured process, i.e. breaking down the problem into stages, evaluating intermediate outputs, and building towards a well-reasoned conclusion.

Video 2: Chain-of-Thought (CoT) Prompting

In the previous demos in this particular module of prompting, we have surveyed zero-shot prompting, few-shot prompting, delimiters, and additional techniques where prompting can be improved. The next advanced technique beyond few-shot prompting is chain-of-thought prompting. Around the start

of 2023 or end of 2022, one paper came which demonstrated the chain-of-thought prompting strategy.

This paper came from a Google lab. What is chain-of-thought prompting? It tells the large language model how to think step by step. Instead of just giving an example that this is how your output should be, it asks it to think step by step.

For example, let us take a one-shot prompt. The question in that chain-of-thought prompting looks like: Roger has five tennis balls, he buys two more cans of tennis balls, and each has three tennis balls. How many tennis balls does he have now? In this one-shot prompt, the standard answer given in the prompt is the answer is 11.

While in chain-of-thought prompting, the answer is not given as 11. The answer is given in an elaborated manner to enable step-by-step thinking. Like Roger has five balls, two cans of three tennis balls each has six balls. So, 6 plus 5 equals 11, the answer is 11. So, instead of just giving a simple answer, a chain-of-thought answer is supplied in this one-shot example. And then the question is asked, and we have seen that this paper illustrates that the standard prompt gives a wrong answer and the chain-of-thought prompt gives a correct answer.

Over these years, in the last 2–3 years since this chain-of-thought prompting strategy has come from Google, this has now become an industry standard in frontier LLMs. Beyond few-shot prompting, most frontier LLMs apply chain-of-thought reasoning in their thinking processes. So, further in this demo, we will try to demonstrate chain-of-thought prompting on ChatGPT.

I have opened the ChatGPT web interface. To display chain-of-thought prompting, I will first put some agenda and context. So, I am putting this context. I work at a SaaS company where we are seeing a spike in customer cancellation. We have data: login frequency, support ticket volumes, plan tier, and last purchase date. I want to use AI to help us understand and manage churn. Can we quickly explain what churn analysis means in simple terms? So, we give this to ChatGPT. Along with this, now I have added additional data. Our five data samples are: customer A has a pro plan, logged in 2x months, four support tickets, purchased last 11 months ago; customer B starter plan, and so on.

I will put this additional context. Now, I will ask ChatGPT to analyse churn risk and suggest actions. I have a very basic prompt. It is not a chain-of-thought prompt. It is a very basic prompt. And let us see how ChatGPT answers. So, it has put customer C in critical risk, customer A in high risk, customer E in high risk, customer D in moderate risk, B in low risk, and so on. We have seen the output of the basic prompt. Now, we are doing the chain-of-thought prompt.

As you can see, I have revised the prompt and said let us do this properly, redo this properly, analyse the churn risk for the five customers I shared, but walk me through step by step. What are the steps? These are the steps that we are specifying. First step is identify key signals. What are the behavioural patterns? What are the warning signs for each customer? Step two, explain the impact. Why does the early signal matter? What does it tell about the intent? Step three is prioritise the risk, rank the customers from highest to lowest with brief logic, and then recommend specific action. As you can see, this prompt is more natural to the way humans think.

Let us see now how ChatGPT reacts to this chain-of-thought prompt. As you can see, we have a much better view of this chain-of-thought prompt here. We know how it is explaining everything in each step. Though the answer order from C, A, E, D, B has changed, the answer in this chain-of-thought is also going closer to explainable AI. The AI is trying to explain the intermediate logic. This gives us a very interesting understanding of chain-of-thought.

We have made the comparison manually. Now let us ask ChatGPT how it will compare the chain-of-thought and the basic one-shot zero-shot prompt in its output. I asked it to compare the first response and the step-by-step response it just gave. What are the three biggest differences? As we have already discussed, ChatGPT also says that explicit reasoning versus implicit judgement, generic playbook versus targeted intervention, and summary prioritisation versus decision-ready prioritisation.

As we have seen, chain-of-thought prompting improves upon the basic one-shot or few-shot prompt. It also adds a sense of explainability because it is trying to reason just like a human would. This also adds to the trust factor in AI.

You can push the boundaries of chain-of-thought by adding more reasoning steps when the task is very complex so that one can mimic the process or thought pattern that goes on in a human brain by providing context in this way. In this manner, one can learn to make use of LLMs in a very effective way to their fullest potential.

Video 3: Tree-of-Thoughts (ToT)

Tree-of-thought prompting is a technique that encourages the AI to explore multiple reasoning paths simultaneously, evaluate each one, and pursue the most promising direction. Rather than moving straight to a single conclusion, the AI considers several approaches before committing to one.

The concept mirrors how experienced professionals actually think through difficult problems. When a senior strategist is evaluating a market entry decision, they do not commit to the first line of reasoning that presents itself. They consider several approaches, test the assumptions behind each, discard the weaker ones, and develop the strongest option further.

Tree-of-thought prompting builds that same discipline into how AI reasons. The structure follows three stages. First, the AI generates multiple branches, which are distinct approaches or solutions to the problem at hand.

Second, it evaluates each branch against defined criteria, assessing feasibility, risk, alignment with objectives, or whatever measure is most relevant. Third, it prunes the weaker branches and develops the strongest one into a full response. This technique is particularly valuable in high-stakes business scenarios where the cost of settling on a poor solution early is significant.

For example, in strategic decision-making, a company may need to evaluate multiple market entry strategies. Using the tree of thought, the AI can generate different approaches such as pricing strategies, partnership models, or geographical expansion options, assess their risks and benefits, and then recommend the most viable path. Similarly, in supply chain optimisation, the model can explore

different sourcing strategies, evaluate cost versus risk trade-offs, and narrow down to the optimal solution.

Another practical use case is in financial analysis. When forecasting cash flow under uncertainty, the AI can simulate different scenarios based on varying assumptions, compare outcomes, and identify the most stable or profitable path. This is far more effective than relying on a single prediction.

Tree-of-thought prompting is not necessary for every task. It is best reserved for complex, open-ended problems where multiple viable approaches exist and where the quality of reasoning matters as much as the final answer. By enabling structured exploration and evaluation, this approach enhances decision quality and supports more robust, business-ready AI outputs.

Video 4: Self-Consistency Prompting

In the previous demo, we have seen how chain-of-thought prompting improves the output and provides a more reliable AI than just few-shot prompting. As a natural extension of chain-of-thought prompting, researchers from Google in 2023 again thought about what could be a natural extension to chain-of-thought. Humans do not just think in a chain of thoughts.

They have multiple frameworks to think, and they have multiple types of reasoning they can make. This is precisely the idea of self-consistency prompting. Both chain-of-thought prompting and self-consistency prompting, it is important to mention, are not restricted to prompting itself.

They can also generalise to pre-training, training the model itself, and even at inference time when the LLM interacts with itself. So, coming to self-consistency prompting, when you have a few-shot prompt given to an LLM in chain-of-thought prompting, it follows step 1, step 2, step 3, step 4. While in self-consistency prompting, we provide parallel reasoning paths. Parallel paths for reasoning.

So, one way is to think in a particular manner, another way is to think in a different manner, and yet another way is to think in another manner. The natural extension of chain-of-thought prompting to self-consistency prompting is that we do not have just one chain of thought. We have multiple parallel chains of thought, and they act as parallel reasoning paths.

For example, as you can see, in 2023, a team from Google again put out this paper in a prestigious conference, and its example reads like this: Janet's ducks lay 16 eggs per day, she eats three for breakfast every morning and bakes muffins for her friends every day with flour. She sells the remainder for \$2 per egg. How much does she make every day? The language model then creates three parallel reasoning paths.

So, the first path is: she has 16 minus 3 minus 4 equal to 9 eggs left. So, she makes 2 into 9, which is \$18. The second reasoning path: this means she sells the remainder, \$2 into 16 minus 4 minus 3, that is \$26, and she eats 3 for breakfast. So, she has 13 left, she makes muffins, so she has 9 eggs left, and so she has 9 eggs, that is \$18. As you can see, different reasoning paths provide two different outputs, and then we label the correct reasoning path and also let the LLM learn the correct method of reasoning on this complex task.

Now, we will display this self-consistency prompting on ChatGPT. I have opened ChatGPT's web interface, and I will set the context that I am a finance manager at a retail company. We are preparing our Q4 revenue forecast, and the board wants a risk assessment. What are the top three factors likely to have a negative impact on Q4? We are in the UK mid-market, mostly in-store, and growing with e-commerce revenue. So, we will put this to ChatGPT. You can see the response that for a UK mid-market store, the highest probability downside risks for Q4 cluster around demand fragility and operational execution.

These are the top three: customer demand volatility, promotional intensity and margin-driven value distortions, and execution risks in omnichannel operations. Now, I paste it and run another prompt, asking ChatGPT to give likelihood as high, medium, low and estimate the revenue impact as a percentage. So, as you can see, ChatGPT has provided estimated revenue impact, 4% to 10%, and so on.

Next, I have again added the second prompt and asked it to re-evaluate, and it has given slightly different numbers, as you have seen in this demo. I will ask ChatGPT the same question again. After asking the same question again and again, I will ask ChatGPT that I have run the same risk assessment three times and got slightly different responses. Based on the three outputs, which factors appeared consistent across all runs, and which appeared only once? What does consistency suggest about confidence level?

So, let us see what ChatGPT responds. Consistent factors that appeared in three runs include customer demand softness, promotional intensity, competitive discounting, omnichannel execution, and inventory fulfilment and availability. These themes were structurally identical in all responses, with only minor variations. The likelihood ratings and impacts were also directionally stable. Factors that appeared only once: none. No additional unique factors were present, which means the output is highly consistent.

This self-consistency prompting significantly improves upon chain-of-thought prompting. But as we combine the learnings of chain-of-thought and self-consistency, we see that we are moving towards explainable and trustworthy AI, where we try to make LLMs mimic human thinking, making them more reliable. Self-consistency prompting also introduces a consistency signal.

These prompting strategies are not limited to prompting alone. They can extend into fine-tuning as well as foundational model training. But it starts with prompting strategy, and it helps us understand how these LLMs are being used and continuously improved day by day.

Video 5: ReAct (Reason + Act)

In this week, we have surveyed the advanced prompting methodologies like chain of thought prompting and self-consistency. And we have also seen that chain of thought prompting is a natural extension above from the few-shot prompting and self-consistency is again a natural way of seeing it. In this demo, we introduce a new extension to self-consistency or an improvement in chain of thought and self-consistency called as the ReAct framework.

The ReAct framework is an abbreviation of Reason plus Act framework. The ReAct framework is introduced by a paper in 2023 jointly with Google and Princeton University. In this framework, they try to compare their Reasoning plus Act framework with the standard prompting, with the chain of thought prompting, with the Act-only prompting on standard datasets.

As you can see on the screen, this is a hotspot QA dataset, a standard benchmark dataset where the ReAct framework is outperforming the Act-only and the Reason-only framework. Similarly, in the ELFWORD dataset, the ReAct framework, Reason plus Action framework, outperforms the Act-only framework. Now further in this demo, we will try to demonstrate this on an LLM.

To demonstrate the ReAct framework or the ReAct prompting in chat-gpt, I will again set the context as in the previous demo. The context is, I am a product manager and my CEO just asked me to quickly assess whether we should enter the AI-powered recruitment software market. I need to understand the competitive landscape, identify market gaps, recommend a positioning angle all in one session.

This is a complex multi-step task, where do I even start? Let us see how chat-gpt answers this. So, it is asking, by first, is the market even worth entering, map the competitive landscape, and then AI first-point solutions, horizontal HR platform embeddings, and so on, then identify real problems, still are the market gaps, synthesise where you can realistically win, and what to tell your CEO. Simple answers.

It has given a very detailed answer to that prompt. Now, after this context setting, I will ask the chat-gpt to give a complete competitive analysis of AI-powered recruitment software, identify gaps, and recommend our position angle. And let us see how it answers.

It shows the market reality, competitive landscape, again value chain breakdown, whether competition is strong versus weakest, core market insights, strategic positioning options, and what not to do, critical and recommended position, and how to articulate to your. My previous prompt was a simple complex prompt, but now we will put a ReAct prompt into here. So, the Reason plus Act approaches start with reason.

Step one, before we look at the specific competitors, do we need to understand about AI-powered recruitment software market to make sound assessment? List 4-5 questions our analysis must answer and explain why each question matters. Now you can reflect on this and think that this is the way usually people think in groups. Let us see how chat-gpt answers this.

It is first giving us the questions. Where is the durable value created in the workflow? Who is the real buyer? Here as you can see, it has given five questions which are synthesised as where, who, how, what, now. These are the reasoning step.

Now we are adding an action step. Action step one, act step one, answer question one or two from our reasoning list. What pain points do recruiters have that existing tools do not solve? And which market segments are currently underserved by AI recruitment? Be specific.

Let us see how chat-gpt answers. So, it has taken some questions and trying to answer that, finding out the pain points, lack of explainability, bias, compliance, and so on. So first we reasoned out the question, then acted on the question, and the third step now here is again the reasoning step based on what we have just learned about the underserved market SMBs, hourly workers, technical

specialists, which of these three segments represent the best market entry opportunities for a new player? Reason before recommending one. Consider market size, competition intensity, and our ability to differentiate. Now you can see a reasoning loop again added after sandwiching two reasoning loops are there and an action loop is there in between.

It has given a summary. Now over the top of the reasoning loop, the latest one, we add one more act step, final action. Based on everything we have reasoned out, write a three paragraph market entry positioning recommendation for entering AI powered recruitment market, targeting technical specialist hiring, include market gap, here addressing our core differentiating angle and first move we should make.

So, instead of one shot prompting, few shot prompting, chain of thought prompting, this delivers a very deep paragraph that is convincing also as a document, main document for this action step. So, we have realised that how ReAct framework improves the reasoning capability of LLM just by asking LLM to act and reason. After this reflection, let us ask this reflection to the chat GPT itself.

In two sentences, what does the ReAct loop procedure produce that the single flat prompt cannot? It produces structured stepwise decomposition of the problem, ensuring key assumptions are explicitly examined before conclusions are drawn. A single prompt typically collapses these steps to surface level biased answer without defensible strategic foundation. This is what in this week we are precisely talking about.

That complex problems can be understood, broken down into more problems, then the ReAct framework, chain of thought prompting, all these advanced prompting strategy can give you a better and better answer, also a trustworthy answer. At the end, we also try to see where we should apply this ReAct framework. Obviously, with our reasoning, natural human reasoning, the problem is complex, the problem is deep, not just complex, not just multidimensional, but also deep.

Let us see what chat GPT thinks about it. Three business case types ReAct would add a significant value. Market entry expansion, complex product strategy, root cause analysis. As we have seen that chat GPT answered the market entry position, complex product strategy, and root cause analysis of business performance issues. These all are both complex and deep problems.

You do not use advanced prompting strategy for day to day tasks. But when the task is complex, new strategies of prompting, advanced prompting strategies, make a lot of difference. And once you try to master all these things, you will make use of LLM both for your business use and the enterprise use. For example, these strategies are not just sitting only at the prompting level on a web interface.

It can go into the fine tuning phase. It can go into the foundational model training phase. It can also go while you choose different models to evaluate which of these prompting strategies are being best done by different models.

Video 6: OpenRouter

In earlier video, we have seen that web interface is not the only interface where you can interact with LLMs. The two other methods are via API keys or you can download the models on your own computer

or a desktop of the server. Today, we will see a demonstration on how you can use API keys and build your own custom system prompt. This is the portal for OpenRouter. In this OpenRouter, we have several models mentioned, listed and we can access all of these models separately. Let us try to see how can we access this model.

First of all, you need to get an API key. You click this API key and then you create an API key. Not to mention that you need to make a personal account on this website. So I will name this API key as, let us say, my chatbot and credit limit I'll just put one and I create this. I will copy this API key. This is the only time you will see this API key, you have to copy it once. So I'll copy this API key. It is in my clipboard now. You can save it somewhere at a place.

Now how to use this API key. There is a portal called make.com in which I have already logged in and tried these API keys. This is the portal for make.com, assuming that you have signed up for a free account here.

You can go and create scenarios. Once you sign in at that portal, you will get some free credits, typically thousand at this point of time and then you get a design board. In this design board, we will click plus. In this plus it will give me many options to link several applications. So since I want to use LLMs via OpenRouter, I will select OpenRouter and I will create a chat completion model. In this OpenRouter, I need to add a connection to OpenRouter.

The connection time is OpenRouter and here I will paste my API key and I will save it. So my OpenRouter is linked and then I will select the model by which I want to run it. So for demonstration purpose, I will select some free model. So let us say I'll select this NVIDIA Nemotron 3 supermodel. You can select any other model, you can select the paid version also, but for the paid version, you need to make sure that OpenRouter has enough funds so that your paid model works. Free model does not require any payment.

There I will add message. Here you can see there is a role and a content. So I will first add a system prompt. System prompt is the prompt that is given to the LLM and it cannot overrule this prompt no matter what comes. So here I will try to assign a role to this prompt in this way. See, assume that you are a senior economist.

You are crisp and to the point. Once I assign this role and I assign the role to the prompt, I can add a user comment. I will add this user comment that suppose what do I want to know? I want to know that tell me the reasons for 2007 global recession.

This is my prompt and the role is already given here. You can add as many details to this content based on the delimiters and the other mechanism that we have gone through in the previous demo. Then I will save it.

So once this OpenRouter is done, then I can add a module again and whatever I want. For example, I want this output to be in my Google Sheets. So I will select the sheets app and I shall select that how the Google Sheets will be updated. So Google Sheets will be updated by adding a row. So now I add a connection to the Google Sheets by signing up. I will sign up here, continue.

Once this sheet is added, I will select what is the sheet name. It will give you, I have a sheet called OpenRouter, so I will select the sheet OpenRouter. Just to show you that in my Google Drive, I have an empty sheet named as OpenRouter.

So once that sheet is there, it will automatically find out this sheet. The sheet name, it will prompt you, you will select the sheet name and it does not contain a header. So I will say no, I need to select the column range where the output will be added.

And I need to select the output. In this output, there are various things that you can do in output. You can output the model name in column A, you can output anything. So, for example, if I want to output in column A model name, in column B I want to output the message and in the message, I want the role and in column C I want the content. So this is the format I want and I'll save it.

Once I save it and just to recall you that what is the prompt that you have given, you have given that tell me the reasons for 2007 global crisis and it will ask this NVIDIA Nemotron 3 this question. After that, once the setup is done, then you can run it once. Before running it, I would like to mention you can add any module to it, either here or previously. What modules you can add, you can add your YouTube channel where the comments are coming and you want to automatically reply to these comments based on the rule you set up.

You can add to Reddit, you can add to Facebook and you can have your business apps like Slack, etc. and you can browse any app here, Google Drive, Slack, Facebook, Microsoft, Gmail, you can add your Gmail, you can add your WhatsApp business, Zoho CRM, Calendly, Google Calendar, anything can be added.

So, for this demo, I will just simply run this command for the LLM and I'll run it once. The scenario is saved and it will take some time to run this command and ask and save the prompt in the Google Sheets.

I will show you in the Google Sheets a little later, but as this is running, if the model is bigger, better and it is a premium model, it will run a little faster. Since we have chosen a free model, it will run a little slower.

Once the running is completed, it will show you that it is successful. Now I will go to the Google Sheet and see the answer. As you can see in column A, it gives you the model name, it gives the role of the assistant that the Nemotron is asking is, the Nemotron is behaving as an assistant and these are the reasons 2007–8 global recession, often called as Great Recession, was primarily triggered by the collapse of US housing market and the four, five or seven reasons are mentioned by this.

So, in this way, you can set up prompt setup and save all this data and output in this manner. Coming back to applications like Make, this is a very, very simple demonstration of how you can use free LLMs via these channels like OpenRouter or make.com.

There are other applications also available to do this, but the power to use this application, how do you use it, relies only on your creativity and your time. So, as you have seen in this demo, that you can use API keys via make.com to connect with your Google Sheet.

As we go further in our module, the usage of LLM and usage of several tools will build up an agentic AI system. So, agentic AI system and the generative AI, agentic AI system is a continuation of generative AI system and one of the key skills apart from the technical skills and the know-hows or how the system work, one of the key delimiter is that how much you can imagine, how much you can implement and try these tools.

This will fire up your imaginations and several business applications will unfold in your organisation. There may be many tasks that you want to automate. How do you want to automate it? This video gives you a visual demo of using several kinds of LLMs into your business domains and business application.

Video 7: Constraint and Guardrail Prompting

In this week, we have seen different prompting mechanisms, specially advanced prompting mechanism. Prompting mechanism like chain of thought prompting, self-consistency and ReAct prompting gives us additional viewpoints how to get leverage of the LLM in the best manner for our complex and deep tasks. But a very important aspect apart from getting the best use of AI is making AI safer.

What do you mean by making AI safer? Safe AI means it should not hallucinate, it should not give harmful content or it should not give incorrect content. We need to put certain guardrails so that AI does not answer or go haywire or give either a harmful answer or a wrong answer. Both will deteriorate the trust and the usability of AI.

In this video, we will talk about how we can set up guardrails using prompting. Though ideally the guardrail should be set up at system prompting, it requires a little bit deeper understanding when you try to deploy a chatbot. But just to illustrate a concept of a guardrail, we will go and see how it is done in ChatGPT.

Just as in the previous demos, we will first set up the context for guardrails. I will paste the prompt. The prompt is setting up the context to develop a question and answering tool where employees can ask about HR policy.

This uses AI and we are asking what are the biggest risks of doing it. Definitely, we are going to paste our actual data and what are the biggest mistakes people make while setting up this AI. So AI is answering in its way.

Key design principles are again not so explicit. We will try to see how those are applied via prompting. So I am pasting the HR policy data and will leave 25% per year for employees must book minimum two weeks in advance maximum five days carried over to the next year sick leave maternity leave expenses and travel economic class only and so on no fully remote roles.

Now let us try to test. We try to add a basic guardrail new rule answers questions only HR provided do not add any information and now answer what is the policy on paternity leave. It does not provide any data on the paternity leave.

Now we refine the prompt for the HR bot and include this strictly answer only HR policy data provided do not provide any outside knowledge if information is not in the provided data explicitly say I do not know. So we are putting an I do not know system prompt never guess or estimate anything that is not explicitly stated test question is also there. And what if I need more than five days of carryover annual leave simply says this is not covered in the policy document please contact HR this is the type of guardrail that we are talking about.

Now let us push the boundaries of guardrail and try to see test question. My manager told me verbally that I can carry 10 days of leave this year as a special exception is that a policy I am sick during this period of annual leave does the sick replace annual leave days simply puts the guardrail does not answer anything it is under covered in the current policy document.

Now let us compare the response of the paternity leave question with no guardrails versus with guardrails in two sentences what has changed and what risk was eliminated without guardrails the model could generate plausible but fabricated paternity leave policy because it is optimised to produce answer even if the data is missing with guardrails it refuses it says I do not know it only sticks to the data.

This is also apart from the guardrails it is also called as the grounding that you are only subscribing only to the ground truth and grounding and guardrails together make our AI more trustworthy. So what are the frameworks of guardrail should be there. It should only so as we have seen in this guardrail example there are a couple of things that we have that guardrail should perform it should say it should only say the information that is available in the document second it should say explicitly no and third it should never guess or estimate anything.

Let us ask AI about this reflection AI says source constraint no answer rule and no inference points. So we have seen that guardrails have significantly improved the trustworthiness usability of AI chatbots along with the advance that you will put behind your chatbot as a system prompt or as an inference strategy these guardrails are must and should be carefully planned and tested before your own automation or chatbot is deployed to employees or to your clients.

Video 8: Final Recap and Prompting Framework

You have covered significant ground this week. Before you move forward, let us bring everything together in a framework that you can carry in your day-to-day work. The techniques in this module fall into three groups, each serving a distinct purpose.

The first group covers reasoning techniques. Chain of thought, few-shot chain of thought, decomposition, least to most prompting, tree of thought all belong here. These are the techniques you reach for when a problem is complex, when the quality of reasoning matters and when a single-step prompt is unlikely to produce a reliable answer.

Use these when you need the AI to think through a problem rather than simply responding to it. The second group covers reliability techniques. This includes self-consistency prompting, where the AI generates multiple responses and converges to the most dependable answer.

Self-verification and reflection, where the AI is prompted to review and challenge its own output. And ReAct prompting, which combines reasoning with action to produce responses that are grounded and traceable. These techniques are most valuable in professional contexts, where outputs inform decisions and are shared with stakeholders or feed into automated workflows.

The third group covers production techniques. Constraint and guardrail prompting, prompt chaining, retrieval-augmented generation and structured output techniques all sit here. These are the approaches that make your prompting repeatable, scalable and safe to deploy across teams and systems.

A legal operations team using prompt chaining to move a contract through drafting, risk flagging and plain language summarisation in a single connected workflow is a strong example of production-level prompting in practise. The combination framework is straightforward. Start with your structured prompt anatomy, role, task, context, constraints and output.

If problem require multi-step reasoning, layer in chain of thought or tree of thought. If the output needs to be reliable enough to act on, apply self-consistency or self-verification. If you need outputs grounded in specific organisational knowledge, bring in retrieval-augmented generation.

If you need scale and consistency across teams, add guardrails and build towards reusable prompt chains. For example, a marketing team generating a campaign strategy can begin with a simple prompt, then add structured reasoning to evaluate alternatives and finally apply output constraints to produce actionable insights. Here is the most important practical tip to take away.

Start simple and layer techniques only when the task demands it. A well-structured zero-shot prompt will handle majority of business tasks effectively. Reserve the advanced techniques for problems where stakes are higher or where the reasoning is more complex or the output needs to meet a higher bar of reliability.

Now it is your turn. Identify one complex task you are currently working on, a strategic brief, a performance analysis, a risk assessment or a stakeholder communication and apply the combination framework to it. Notice when layering a reasoning or reliability technique changes the quality of what you get back.

The prompting toolkit you have built across this module is one of the most transferable skills you can develop as AI becomes more central to how organisations operate. Use it deliberately, refine it with experience and it will continue to deliver.

Video 9: Final Thoughts

We have reached the end of this module and it is worth taking a moment to recognise how far you have come. When you started, the goal was to move beyond basic prompting into techniques that match the complexity of real business problems. You have done that from chain of thought reasoning to tree of thought exploration, from self-verification to prompt chaining, from guardrail design to retrieval augmented generation.

You now have a toolkit that most professionals have not yet built. But a toolkit only creates value when it is used. The techniques you have learned through demonstrations are not simply academic exercises.

They are practical instruments designed for the kinds of problems you face every day. The problems that require structured thinking, reliable outputs and confidence to act on what AI produces. What separates advanced prompting from basic prompting is not complexity for its own sake.

It is intentionality. Knowing which technique to reach for and when to layer approaches and when a simple prompt is enough. That judgement is what you have developed here. As AI becomes more and more deeply embedded into how organisations think, decide and operate, that judgement will matter more, not less. Carry it forward, apply it consistently and let it compound with every task you take on.